

COMPLETED WORK SUMMARY

Project Completion Summary

Overview

The objective of this project was to transform raw Airbnb marketplace data into meaningful business intelligence through the application of data engineering and exploratory analytics techniques.

The project successfully completed the majority of the core analytical workflow, resulting in a comprehensive assessment of pricing behaviour, demand patterns, host activity, and geographic market characteristics.

Data Engineering Activities Completed

The following engineering tasks were completed successfully:

- Dataset profiling and structure analysis
- Missing value assessment
- Data quality validation
- Data cleaning and standardization
- Date transformation and temporal processing
- Listing Master Dataset creation
- Occupancy rate calculation
- Revenue estimation
- Review frequency calculation
- Host tenure calculation
- Host portfolio classification
- Dimensional data modeling

These activities created a robust analytical foundation capable of supporting advanced business analysis.

Exploratory Data Analysis Completed

Extensive exploratory analysis was conducted across multiple business dimensions.

Completed analyses included:

- Overall pricing analysis
- Room type pricing comparisons
- Neighbourhood pricing analysis
- Geographic listing distribution
- Geographic pricing patterns
- Seasonal availability trends
- Review volume trends
- Host tenure analysis
- Host segmentation analysis
- Supply concentration analysis
- Demand-side analysis

More than twenty analytical visualizations were developed to support findings and business interpretations.

Business Intelligence Outcomes

The project generated several practical business insights, including:

- Identification of premium neighbourhoods
- Understanding of host concentration patterns
- Evaluation of supply-side market structure
- Discovery of seasonal demand behaviour
- Analysis of pricing drivers
- Identification of high-demand listing characteristics

These findings formed the basis for actionable business recommendations.

Deliverables Produced

The project successfully produced:

- Source code notebook

- Analytical dataset
- Data model documentation
- Professional report
- Business recommendations
- Supporting project documentation

Conclusion

Overall, the project successfully achieved its primary objective of converting raw Airbnb data into actionable business intelligence while demonstrating practical data engineering and analytical capabilities.